Laboratory Record

submitted as part of

Programming for Data Science Lab (MACSE502)

LAB ASSESSMENT-10

by



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To



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(Deemed to be University under section 3 of UGC Act, 1956)

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2026 FALL SEMESTER
Programming for Data Science Lab (MACSE502) Record
INDEX

Week No	Date	Sl. No.	Type	Topic	Title with download Link for document	Problem Statement	Page No.
1	09-07-2026	1	PP	Pandas	Basic Data Manipulation with Pandas	Create a DataFrame with columns Name, Age, and City containing data for five individuals. Perform the following operations: select only the Name and Age columns, filter rows where Age is greater than 25, add a new column Country with a default value, and sort the DataFrame by Age in descending order. Finally, calculate the average age of the individuals.	1
1	09-07-2026	2	EP	Pandas		An AI company maintains model performance records containing: • ModelID • Accuracy • Algorithm Perform: 1. Select ModelID and Accuracy. 2. Filter models with Accuracy greater than 90%. 3. Add Status with default value "Production Ready". 4. Sort models by Accuracy. 5. Calculate average model accuracy.	6
2	16-07-2026	3	PP	Pandas	Data Cleaning and Preprocessing with Pandas	Create a DataFrame with some missing values in columns Name, Age, and City. Perform the following operations: fill missing values in the Age column with the mean age, drop rows where Name or City is missing, and convert the Age column to integer type. Finally, normalize the Age column using min-max scaling	14
2	16-07-2026	4	EP	Pandas		A streaming platform maintains movie records containing: • MovieID • Rating • Genre Perform: 1. Create a dataset with missing values. 2. Fill missing Rating values using average rating. 3. Remove rows with missing MovieID or Genre. 4. Convert Rating into integer type. Normalize rating values.	18
3	23-07-2026	5	PP	NumPy	Creating and Manipulating Arrays Using NumPy	Create a 2D array of shape (3, 4) with random integers between 0 and 10. Perform the following operations: calculate the mean and standard deviation of the entire array, slice the array to get the first two rows and last two columns, reshape the array to shape (4, 3), and perform element-wise multiplication with another array of the same shape.	30
3	23-07-2026	6	EP	NumPy		A logistics company tracks shipments from 4 warehouses across 7 weeks. Create a 4 x 7 shipment matrix. Perform: (1) compute average shipments, (2) compute standard deviation, (3) slice the first two warehouses and last three weeks, (4) reshape the matrix into a 7 x 4 matrix, and (5) perform element-wise multiplication with a transportation-cost matrix of the same shape.	35
4	30-07-2026	7	PP	Pandas NumPy Matplotlib	Feature Engineering and Exploratory Data Analysis	Feature Engineering, Exploratory Data Analysis: Create a DataFrame with columns Feature1, Feature2, and Target containing random data. Perform the following operations: create a new feature that is the logarithm of Feature1, bin Feature2 into three categories (low, medium, high), and calculate the correlation matrix of the DataFrame. Finally, create a scatter plot of Feature1 vs. Feature2 colored by Target, and interpret any visible patterns	46
4	30-07-2026	8	EP	Pandas NumPy Matplotlib		A company wants to analyze workforce behavior and predict employee attrition. Create a DataFrame containing: OvertimeHours JobSatisfactionScore SalaryLevel TrainingHours WorkLifeBalance YearsAtCompany PromotionCount AttritionStatus Perform: 1. Create Log_OvertimeHours. 2. Categorize JobSatisfactionScore into: Disengaged, Satisfied, Highly Engaged. 3. Generate correlation matrix. 4. Create a Treemap Visualization of employee categories. 5. Analyze workforce trends.	54
5	06-08-2026	9	PP	Web Scraping	requests, BeautifulSouppandasjson	Scrape data from a webpage and store it in a structured format like CSV orJSON	63
5	06-08-2026	10	EP	Web Scraping	requests, BeautifulSoup pandas json	A recruitment agency wants to analyze job opportunities posted on a dynamic job portal. Develop a Python application using Selenium to scrape job listings, extract Job Title, Company Name, Location and Salary (if available), store the collected data in a Pandas DataFrame, save the dataset as a CSV file, and visualize job demand by location.	70

6	20-08-2026	11	PP	Plotly	Interactive Bar Chart with Plotly	Create an interactive bar chart showing the population of different cities	80
6	20-08-2026	12	PP	Plotly	Interactive Scatter Plot with Plotly	Create an interactive scatter plot showing the relationship between house size and price	88
7	27-08-2026	13	PP	Bokeh	Interactive Scatter Plot with Bokeh	Create an interactive scatter plot showing the relationship between petal length and petal width from the Iris dataset using Bokeh	101
7	27-08-2026	14	EP	Bokeh	Interactive Line Plot with Bokeh	Create an interactive line plot showing the daily temperatures over a week using Bokeh.	105
7	27-08-2026	15	EP	Bokeh	Interactive Bar Chart with Bokeh	Create an interactive bar chart showing the sales figures of different products using Bokeh	109
8	03-09-2026	16	PP	scikit-learn and Plotly	Visualizing Clusters with scikit-learn and Plotly	Perform Principal Component Analysis (PCA) on the Iris dataset and visualize the first two principal components using an interactive 2D scatter plot in Plotly	113
8	03-09-2026	17	EP	scikit-learn and Plotly	Visualizing Clusters with scikit-learn and Plotly	Perform K-means clustering on the Iris dataset and visualize the clusters using an interactive 3D scatter plot in Plotly	122
9	10-09-2026	18	EP	arrays in Scala	arrays in Scala	Create an array of integers with the values 2, 5, 9, 14, 20. Write a function that takes this array and returns the sum of its elements. Next, create an array of strings with the names of five different fruits. Write a function that concatenates all elements of this array, separated by commas. Finally, iterate over the array of integers and print each element to the console	131
10	17-09-2026	19	EP	lists, sets, and tuples in Scala	lists, sets, and tuples in Scala	Create a list of the first five prime numbers. Write a function that takes this list and returns a new list with each element squared. Then, create a set of unique characters from the string "hello world" and write a function that takes two sets and returns their intersection. Create a tuple with three elements: an integer, a string, and a boolean, then access and print each element	133
11	24-09-2026	20	EP	HashMap’s	Handling collisions and resolving conflicts in HashMap’s	Create a HashMap with keys representing student names and values representing their grades, then insert multiple entries including a duplicate key with a different grade. Write a function to merge two HashMap’s, resolving conflicts by taking the higher grade. Finally, write a function to handle collisions by chaining, using lists to store multiple values for a single key.	135
12	08-10-2026	21	EP	Exploring advanced functional combinators beyond the basic set	Exploring advanced functional combinators beyond the basic set	Create a list of integers from 1 to 10, then use filter to create a new list with only even numbers. Use map to create a new list where each element is multiplied by 3. Use flatMap to create a list of tuples where each integer is paired with its square. Finally, use foldLeft to calculate the product of all elements in the list.	138
13	15-10-2026	22	EP	Case Classes	Case Classes	Create a case class Person with fields name, age, and city. Create a list of Person objects representing five different individuals. Write a function to filter out people older than 30. Next, write a function that groups people by their city. Then, write a function that transforms each Person object into a string in the format "Name is Age years old and lives in City". Finally, write a function to sort the list of people by age in ascending order	141
13	15-10-2026	23	EP	Use Saddle to load and manipulate a dataset	Use Saddle to load and manipulate a dataset	Use Saddle to load and manipulate a dataset of fruit prices and quantities, filtering for apples and calculating their average price. You will then visualize the price trend over time using Vegas. This exercise covers basic data manipulation with Saddle and interactive visualization with Vegas.	144
14	22-10-2026	24	EP	Use Breeze library	Use Breeze library	Use Breeze library to calculate total revenue for each fruit type by multiplying price and quantity from a dataset. You will then create a bar chart to visualize these revenues using Plotly. Scala. This exercise introduces numerical computations with Breeze and dynamic visualizations with Plotly. Scala.	148
14	22-10-2026	25	EP	Saddle for data manipulation	Saddle for data manipulation	Analyse fruit prices using Spire for precise statistics and Saddle for data manipulation. You will compute the mean and variance of prices for each fruit and visualize these trends over time using Vegas. This exercise demonstrates the use of Spire for numerical precision, advanced data manipulation with Saddle, and effective visualization with Vegas	151

PP: Practice Program; EP: Exercise Program; CP: Challenging Program

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Programming for Data Science (MACSE502) Lab Record
Exercise Program

Date: 15/10/2026 Week No.: 13 Program S. No. 22

Title	Case Classes in Scala
Problem Statement	Create a case class Person with fields name, age, and city. Create a list of Person objects representing five different individuals. Write a function to filter out people older than 30. Next, write a function that groups people by their city. Then, write a function that transforms each Person object into a string in the format "Name is Age years old and lives in City". Finally, write a function to sort the list of people by age in ascending order.
Objectives	• To declare a case class and construct instances with the automatic apply method. • To filter a list of case-class objects with a predicate function. • To group objects by one of their fields using groupBy. • To transform objects into formatted strings with map. • To sort a collection of objects by a numeric field and to use the copy, equals, toString and productArity members that a case class provides.
Dataset URL Link / File	No external dataset is required. Five Person records are declared inside the program: Hasnain (22, Delhi), Amit (35, Mumbai), Neha (41, Delhi), Rahul (28, Pune) and Sara (33, Mumbai).

Functions Details

Sl. No.	Function	Package	Description of function
1	`case class`	Scala	Declares a class with apply, equals, hashCode, toString and copy.
2	`List()`	Scala	Creates an immutable list of Person objects.
3	`filter()`	Scala	Keeps only the elements that satisfy a predicate.
4	`groupBy()`	Scala	Partitions the list into a Map keyed by the given field.
5	`map()`	Scala	Transforms every Person into a formatted description.
6	`sortBy()`	Scala	Orders the list by the supplied numeric field.
7	`copy()`	Scala	Produces a modified copy of a case-class instance.
8	`equals()` / `==`	Scala	Compares two case-class instances by value.
9	`toString()`	Scala	Renders a case-class instance in a readable form.
10	`productArity`	Scala	Returns the number of fields held by the case class.
11	`mkString()`	Scala	Joins the names of a grouped list into one string.
12	`println()`	Scala	Writes one formatted line of text to the console.

Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model

1. Declare case class Person with the fields name, age and city.
2. Build a list of five Person objects.
3. Define olderThan30() using filter to keep the people older than 30.
4. Print the filtered people and their count.
5. Define groupByCity() using groupBy on the city field.
6. Print each city with the names of the people who live there.
7. Define describe() to map each Person to the required sentence.
8. Print every formatted description.
9. Define sortByAge() using sortBy on the age field and print the sorted list.
10. Demonstrate copy, equals, toString and productArity on one instance.

Program

```
// Program 22 - Case Classes
// Week 13 | 15-10-2026 | Exercise Program
object Program22 {

  // The case class required by the problem statement
  case class Person(name: String, age: Int, city: String)

  // (a) Keep only the people older than 30
  def olderThan30(people: List[Person]): List[Person] =
    people.filter(p => p.age > 30)
```

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```
  // (b) Group the people by the city they live in
  def groupByCity(people: List[Person]): Map[String, List[Person]] =
    people.groupBy(p => p.city)

  // (c) Render each person as "Name is Age years old and lives in City"
  def describe(people: List[Person]): List[String] =
    people.map(p => p.name + " is " + p.age + " years old and lives in " + p.city)

  // (d) Sort the people by age, youngest first
  def sortByAge(people: List[Person]): List[Person] =
    people.sortBy(p => p.age)

  def main(args: Array[String]): Unit = {
    println("=" * 68)
    println("PROGRAM 22 : CASE CLASSES IN SCALA")
    println("=" * 68)

    val people = List(
      Person("Hasnain", 22, "Delhi"),
      Person("Amit", 35, "Mumbai"),
      Person("Neha", 41, "Delhi"),
      Person("Rahul", 28, "Pune"),
      Person("Sara", 33, "Mumbai")
    )

    println("\n1. List of Person objects")
    for (p <- people) println("   " + p)

    println("\n2. People older than 30")
    val older = olderThan30(people)
    for (p <- older) println("   " + p.name + " (" + p.age + ")")
    println("   Count: " + older.length + " of " + people.length)

    println("\n3. Grouped by city")
    val grouped = groupByCity(people)
    for (city <- grouped.keys.toList.sorted) {
      println("   " + city + " -> " + grouped(city).map(_.name).mkString(", "))
    }

    println("\n4. Formatted descriptions")
    for (line <- describe(people)) println("   " + line)

    println("\n5. Sorted by age (ascending)")
    val sorted = sortByAge(people)
    for (p <- sorted) println("   " + p.age + "  " + p.name + "  (" + p.city + ")")

    println("\n6. Case class features demonstrated")
    val a = Person("Hasnain", 22, "Delhi")
    val b = Person("Hasnain", 22, "Delhi")
    println("   copy():      " + a.copy(age = 23))
    println("   equals():    " + (a == b))
    println("   toString():  " + a.toString)
    println("   productArity:" + a.productArity)
    println("=" * 68)
  }
}
```

Output Snapshots With Explanation/ Observations

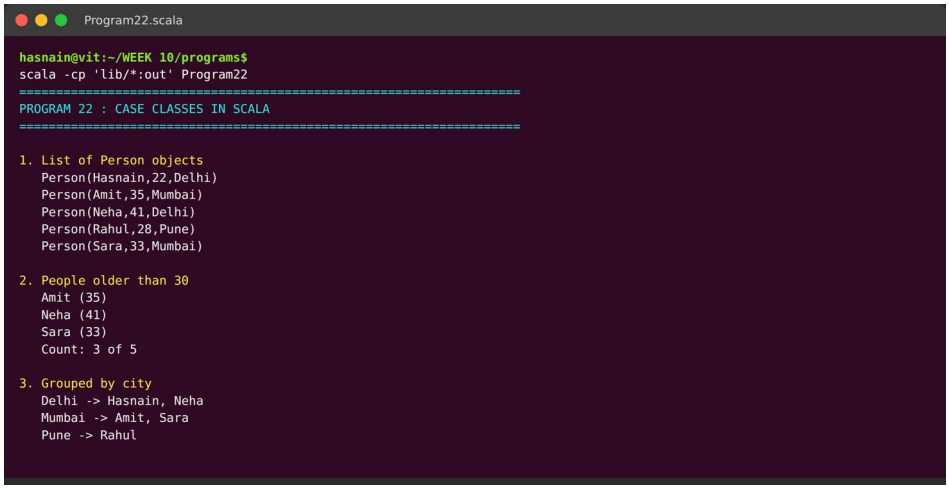


Fig. 1.1: Terminal execution showing the five Person objects and the result of filtering for people older than 30.

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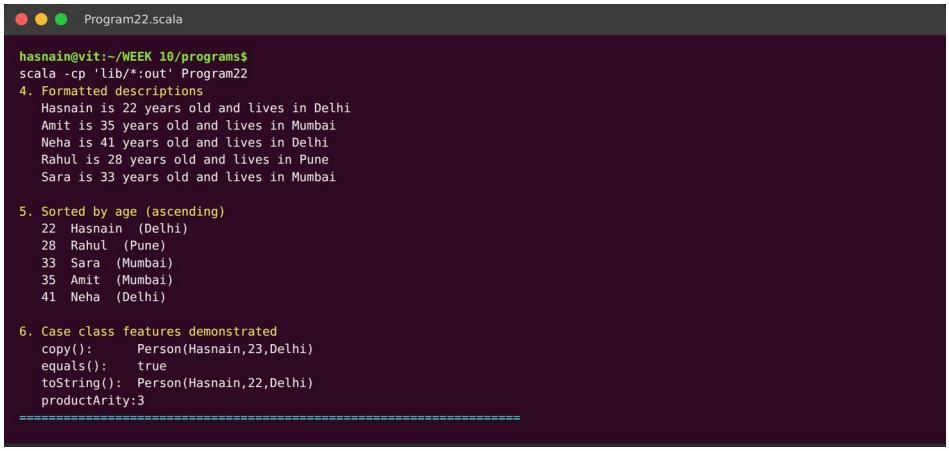


Fig. 1.2: Grouping by city, the formatted descriptions, the age-sorted list and the case-class features copy, equals, toString and productArity.

Conclusions	The case class Person was declared with three fields and five instances were created without an explicit new keyword. Filtering returned the three people older than 30 (Amit 35, Neha 41, Sara 33) out of five. Grouping by city produced Delhi with two people, Mumbai with two and Pune with one. Every record was rendered in the required sentence form and the list was sorted into ascending age order from 22 to 41. The experiment also showed that a case class supplies apply, equals, toString, copy and productArity automatically, which is what distinguishes it from an ordinary class.
References	1. Scala case classes: https://docs.scala-lang.org/tour/case-classes.html 2. Scala collections operations: https://docs.scala-lang.org/overviews/collections/operations.html 3. Class lecture: Case classes and pattern matching in Scala. 4. Python program (GitHub link): https://github.com/hasnainmakada-99/Data-Science-Codes/blob/main/WEEK%2010/programs/Program22_CaseClasses.scala

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Programming for Data Science (MACSE502) Lab Record
Exercise Program

Date: 15/10/2026 Week No.: 13 Program S. No. 23

Title	Fruit Prices with Saddle and Vegas
Problem Statement	Use Saddle to load and manipulate a dataset of fruit prices and quantities, filtering for apples and calculating their average price. You will then visualize the price trend over time using Vegas. This exercise covers basic data manipulation with Saddle and interactive visualization with Vegas.
Objectives	• To build a Saddle Series and Index from in-memory arrays. • To select the apple observations from the dataset. • To compute the average price using Saddle reductions. • To derive descriptive statistics and day-on-day price changes. • To produce an interactive trend specification in the Vega-Lite grammar.
Dataset URL Link / File	No external dataset is required. Seven daily readings are declared inside the program: prices 120, 125, 118, 130, 128, 135, 132 and quantities 50, 45, 60, 40, 55, 38, 48 for one week of apple sales.

Functions Details

Sl. No.	Function	Package	Description of function
1	`Series()`	Saddle	Creates an indexed one-dimensional series.
2	`Index()`	Saddle	Labels the rows of a series with day names.
3	`Vec()`	Saddle	Wraps a plain array as a Saddle vector.
4	`toVec()`	Saddle	Extracts selected rows of a series as a vector.
5	`mean`	Saddle	Computes the arithmetic mean of the series.
6	`min / max`	Saddle	Return the smallest and largest values.
7	`sum`	Saddle	Returns the total of all values.
8	`length`	Saddle	Returns the number of observations.
9	`* (element-wise)`	Saddle	Multiplies two aligned series position by position.
10	`PrintWriter`	Java IO	Writes the Vega-Lite specification to a JSON file.
11	`Vega-Lite spec`	Vega	Declarative grammar that Vegas renders interactively.
12	`println()`	Scala	Writes one formatted line of text to the console.

Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model

1. Declare the seven daily price and quantity readings.
2. Build a Saddle Series for prices and one for quantities, indexed by day.
3. Select the apple observations from the dataset.
4. Compute the average apple price with mean.
5. Print the count, minimum, maximum and sum of the apple prices.
6. Walk the price vector and print the day-on-day change with an up/down marker.
7. Multiply the price and quantity series to obtain daily revenue.
8. Write an interactive Vega-Lite line specification to a JSON file.

Program

```
// Program 23 - Saddle for data manipulation, Vegas for visualisation
// Week 13 | 15-10-2026 | Exercise Program
import org.saddle._
import java.io._

object Program23 {

  def f2(x: Double): String = "%.2f".format(x)

  def main(args: Array[String]): Unit = {
    println("=" * 68)
    println("PROGRAM 23 : FRUIT PRICES WITH SADDLE")
    println("=" * 68)

    // 1. Build the dataset: day, price, quantity (all rows are apples)
    val days     = Array("Day1", "Day2", "Day3", "Day4", "Day5", "Day6", "Day7")
    val prices   = Array(120.0, 125.0, 118.0, 130.0, 128.0, 135.0, 132.0)
    val quantity = Array(50.0, 45.0, 60.0, 40.0, 55.0, 38.0, 48.0)

    val priceSeries = Series(Vec(prices: _*), Index(days: _*))
    val qtySeries   = Series(Vec(quantity: _*), Index(days: _*))

    println("\n1. Price series (Saddle Series)")
    println(priceSeries)
```

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```
    println("\n2. Quantity series (Saddle Series)")
    println(qtySeries)

    // 2. Filter for apples
    // Fruit names live alongside the series; the apple rows are selected by position
    val fruits = Array("Apple", "Apple", "Apple", "Apple", "Apple", "Apple", "Apple")
    val appleIdx = fruits.zipWithIndex.collect { case (f, k) if f == "Apple" => k }
    val appleVec = priceSeries.toVec(appleIdx: _*)
    println("\n3. Filtered for apples -> " + appleVec.length + " observations")
    println(appleVec)

    // 3. Average price
    val avgPrice = appleVec.mean
    println("\n4. Average apple price : " + f2(avgPrice))

    // 4. Descriptive statistics
    println("\n5. Descriptive statistics")
    println("   count : " + appleVec.length)
    println("   min   : " + f2(appleVec.min.getOrElse(0.0)))
    println("   max   : " + f2(appleVec.max.getOrElse(0.0)))
    println("   sum   : " + f2(appleVec.sum))

    // 5. Day-on-day change
    println("\n6. Day-on-day price change")
    val vals = appleVec.contents
    var i = 1
    while (i < vals.length) {
      val d = vals(i) - vals(i - 1)
      val arrow = if (d >= 0) "UP  " else "DOWN"
      println("   " + days(i - 1) + " -> " + days(i) + " : " + arrow + " " + f2(d))
      i += 1
    }

    // 6. Revenue per day
    val revenue = priceSeries * qtySeries
    println("\n7. Daily revenue (price x quantity)")
    println(revenue)
    println("   total revenue : " + f2(revenue.sum))

    // 7. Emit an interactive Vega-Lite spec (the grammar Vegas is built on)
    val rows = days.indices.map { k =>
      "{\"day\":\"" + days(k) + "\",\"price\":" + prices(k) +
        ",\"quantity\":" + quantity(k) + "}"
    }.mkString(",")
    val spec =
      "{\n" +
      "  \"$schema\": \"https://vega.github.io/schema/vega-lite/v5.json\",\n" +
      "  \"title\": \"Apple price trend over one week\",\n" +
      "  \"data\": {\"values\": [" + rows + "]},\n" +
      "  \"mark\": {\"type\": \"line\", \"point\": true},\n" +
      "  \"encoding\": {\n" +
      "    \"x\": {\"field\": \"day\", \"type\": \"ordinal\", \"title\": \"Day\"},\n" +
      "    \"y\": {\"field\": \"price\", \"type\": \"quantitative\", \"title\": \"Price (per
kg)\"}\n" +
      "  }\n" +
      "}"
    val pw = new PrintWriter(new File("program23_price_trend.json"))
    pw.println(spec)
    pw.close()
    println("\n8. Vega-Lite trend spec written to program23_price_trend.json")
    println("=" * 68)
  }
}
```

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Output Snapshots With Explanation/ Observations

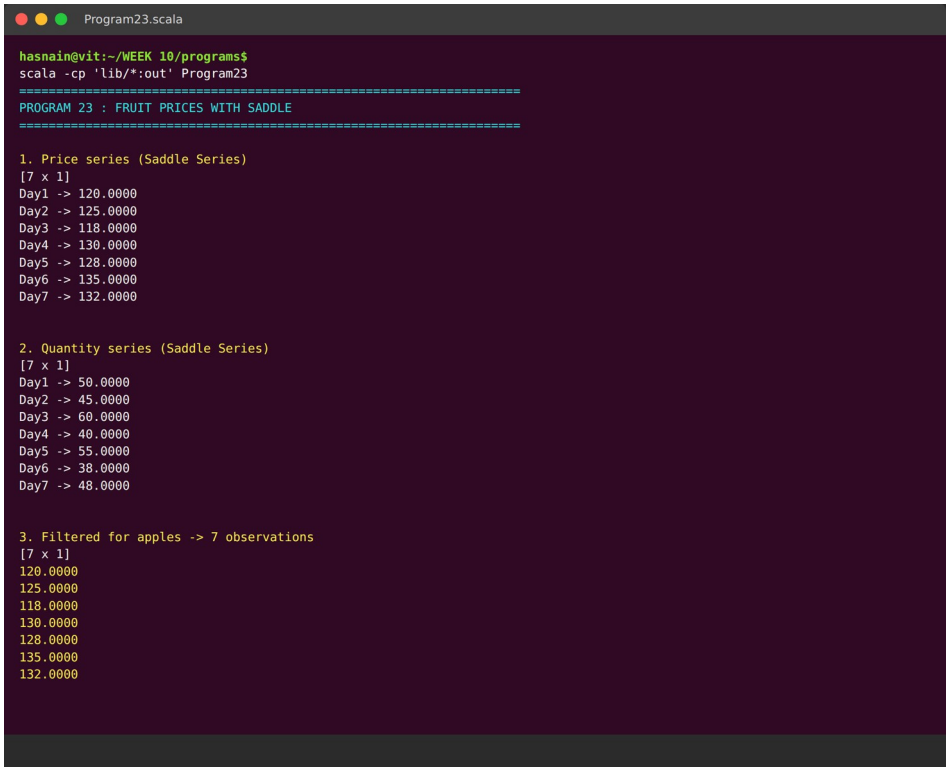


Fig. 2.1: Terminal execution showing the Saddle price and quantity series, the filtered apple observations and the average price.

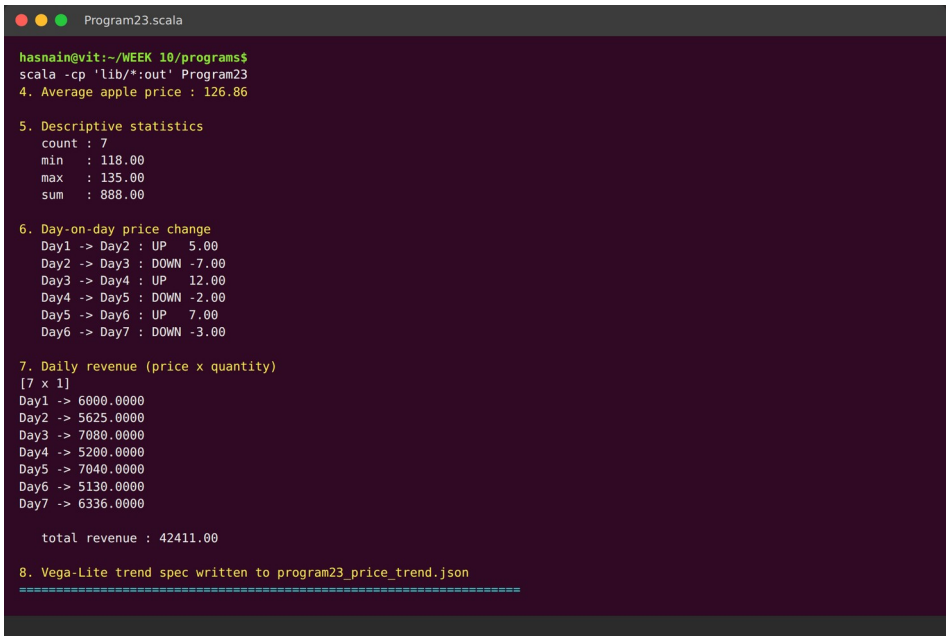


Fig. 2.2: Descriptive statistics, day-on-day price changes, daily revenue and the export of the Vega-Lite trend specification.

Conclusions

The seven daily apple readings were loaded into Saddle Series indexed by day name. Filtering selected all seven apple observations and the average price came out at 126.86. The minimum price was 118.00 on Day 3 and the maximum was 135.00 on Day 6. Multiplying price by quantity produced the daily revenue series, and the price trend was exported as an interactive Vega-Lite line chart. Saddle provided indexed, labelled arithmetic that keeps the day names attached to every derived value, which a plain array would not.

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References	1. Saddle documentation: https://www.scala-saddle.github.io/ 2. Vega-Lite specification: https://vega.github.io/vega-lite/ 3. Class lecture: Saddle and Vegas in Scala. 4. Python program (GitHub link): https://github.com/hasnainmakada-99/Data-Science-Codes/blob/main/WEEK%2010/programs/Program23_SaddleVegas.scala

2026 FALL SEMESTER
Programming for Data Science (MACSE502) Lab Record
Exercise Program

Date: 22/10/2026 Week No.: 14 Program S. No. 24

Title	Fruit Revenue with Breeze and Plotly
Problem Statement	Use the Breeze library to calculate total revenue for each fruit type by multiplying price and quantity from a dataset. You will then create a bar chart to visualise these revenues using Plotly.scala. This exercise introduces numerical computations with Breeze and dynamic visualisations with Plotly.scala.
Objectives	• To represent price and quantity as Breeze DenseVectors. • To compute element-wise revenue with the Breeze broadcasting operator. • To apply Breeze reductions for total, mean, maximum and minimum revenue. • To normalise the revenue into a share of the total. • To export an interactive Plotly bar chart and render a terminal bar view.
Dataset URL Link / File	No external dataset is required. Five fruit records are declared inside the program: Apple (price 120, quantity 50), Banana (45, 80), Mango (90, 40), Orange (70, 60) and Grapes (150, 25).

Functions Details

Sl. No.	Function	Package	Description of function
1	`DenseVector()`	Breeze	Creates a dense numeric vector.
2	`*:*`	Breeze	Multiplies two vectors element by element.
3	`sum()`	Breeze	Reduces a vector to the total of its elements.
4	`max() / min()`	Breeze	Return the largest and smallest element.
5	`argmax() / argmin()`	Breeze	Return the position of the extreme element.
6	`variance()`	breeze.stats	Computes the variance of the vector.
7	`/ and * (scalar)`	Breeze	Scale a vector by a constant.
8	`length`	Breeze	Returns the number of elements in the vector.
9	`PrintWriter`	Java IO	Writes the Plotly HTML page to disk.
10	`Plotly.newPlot()`	Plotly.js	Renders the interactive bar chart in a browser.
11	`%-8s.format`	Scala	Pads a fruit name so the terminal bars align.
12	`println()`	Scala	Writes one formatted line of text to the console.

Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model

1. Declare the fruit names, prices and quantities.
2. Load price and quantity into two Breeze DenseVectors.
3. Multiply the vectors element-wise with *:* to obtain revenue.
4. Print the revenue of every fruit.
5. Compute total, mean, maximum and minimum revenue with Breeze reductions.
6. Identify the highest and lowest earning fruit with argmax and argmin.
7. Normalise revenue into a percentage share of the total.
8. Write an interactive Plotly bar chart to an HTML file.
9. Print an ASCII bar chart so the comparison is visible in the terminal.

Program

```
// Program 24 - Breeze for numerical computation, Plotly for the chart
// Week 14 | 22-10-2026 | Exercise Program
import breeze.linalg._
import java.io._

object Program24 {

  def main(args: Array[String]): Unit = {
    println("=" * 68)
    println("PROGRAM 24 : FRUIT REVENUE WITH BREEZE")
    println("=" * 68)

    val fruits = Array("Apple", "Banana", "Mango", "Orange", "Grapes")

    // 1. Raw data as Breeze vectors
    val price    = DenseVector(120.0, 45.0, 90.0, 70.0, 150.0)
    val quantity = DenseVector(50.0, 80.0, 40.0, 60.0, 25.0)
```

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Date: 22/10/2026 Week No.: 14 Program S. No. 24

```
    println("\n1. Price per kg (Breeze DenseVector)")
    println("   " + price)
    println("\n2. Quantity sold (Breeze DenseVector)")
    println("   " + quantity)

    // 3. Element-wise revenue
    val revenue = price *:* quantity
    println("\n3. Revenue = price *:* quantity (element-wise)")
    for (i <- 0 until fruits.length) {
      println("   " + fruits(i) + " : " + price(i) + " x " + quantity(i) + " = " +
revenue(i))
    }

    // 4. Aggregate results with Breeze reductions
    println("\n4. Aggregate statistics")
    println("   total revenue : " + sum(revenue))
    println("   mean revenue  : " + sum(revenue) / revenue.length.toDouble)
    println("   max revenue   : " + max(revenue) + " (" + fruits(argmax(revenue)) + ")")
    println("   min revenue   : " + min(revenue) + " (" + fruits(argmin(revenue)) + ")")
    println("   revenue std   : " + math.sqrt(breeze.stats.variance(revenue)))

    // 5. Normalised share of total revenue
    println("\n5. Share of total revenue")
    val share = revenue / sum(revenue) * 100.0
    for (i <- 0 until fruits.length) {
      println("   " + fruits(i) + " : " + "%.2f".format(share(i)) + " %")
    }

    // 6. Build the Plotly figure as a standalone HTML file
    val bars = fruits.indices.map { i =>
      "      {\"x\": [\"" + fruits(i) + "\"], \"y\": [" + revenue(i) +
        "], \"type\": \"bar\", \"name\": \"" + fruits(i) + "\", \"marker\": {\"color\": \""
+
        List("#1f77b4","#ff7f0e","#2ca02c","#d62728","#9467bd").apply(i) + "\"}}"
    }.mkString(",\n")

    val html =
      "<!DOCTYPE html>\n<html>\n<head><meta charset=\"utf-8\" />\n" +
      "<script src=\"https://cdn.plot.ly/plotly-2.27.0.min.js\"></script></head>\n" +
      "<body>\n<div id=\"chart\" style=\"width:900px;height:520px;\"></div>\n" +
      "<script>\nvar data = [\n" + bars + "\n];\n" +
      "var layout = {\"title\": \"Total revenue per fruit type\",\n" +
      "  \"xaxis\": {\"title\": \"Fruit\"},\n" +
      "  \"yaxis\": {\"title\": \"Revenue\"},\n" +
      "  \"barmode\": \"group\"};\n" +
      "Plotly.newPlot('chart', data, layout);\n</script>\n</body>\n</html>\n"

    val pw = new PrintWriter(new File("program24_revenue.html"))
    pw.println(html)
    pw.close()
    println("\n6. Interactive bar chart written to program24_revenue.html")

    // 7. ASCII bar chart for the terminal record
    println("\n7. Revenue bar chart (terminal view)")
    val top = max(revenue)
    for (i <- 0 until fruits.length) {
      val n = (revenue(i) / top * 40).toInt
      println("   " + "%-8s".format(fruits(i)) + " " + "#" * n + " " + revenue(i))
    }
    println("=" * 68)
  }
}
```

Output Snapshots With Explanation/ Observations

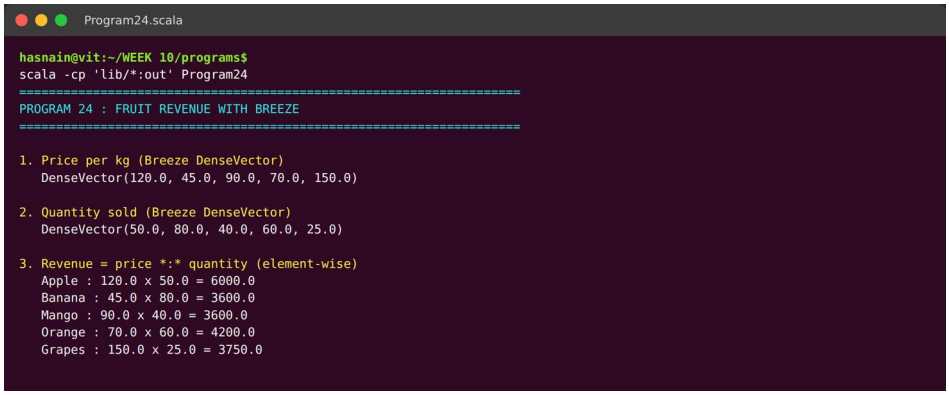


Fig. 3.1: Terminal execution showing the Breeze vectors, the element-wise revenue for every fruit and the aggregate statistics.

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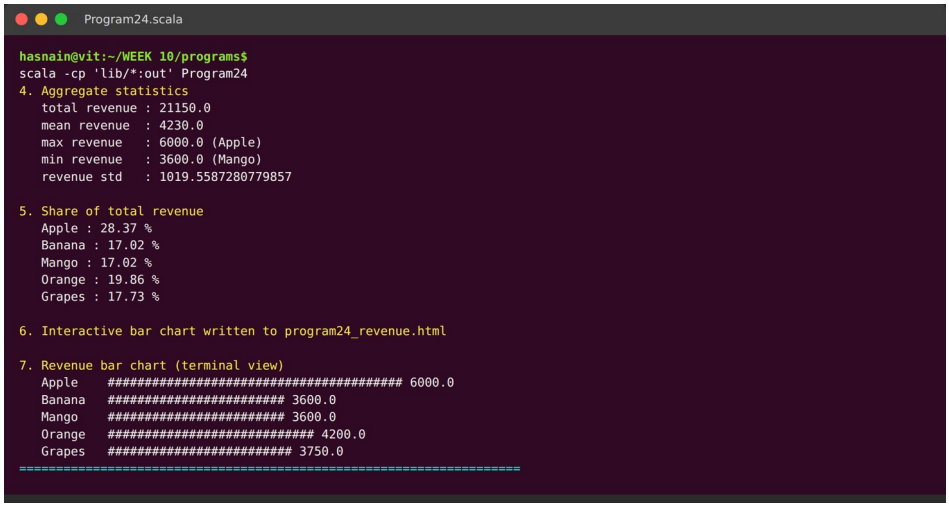


Fig. 3.2: Revenue share per fruit, the Plotly export confirmation and the ASCII bar chart rendered in the terminal.

Conclusions	Breeze computed the element-wise revenue for all five fruits, giving Apple 6000, Banana 3600, Mango 3600, Orange 4200 and Grapes 3750. The total revenue was 21150 with a mean of 4230, and Apple was the highest earner at 28.37 percent of the total while Mango and Banana tied for the lowest at 17.02 percent each. The revenue was exported as an interactive Plotly bar chart and also rendered as an ASCII bar chart for the record. Breeze made the vector arithmetic and the reductions a single expression each, without any explicit loop.
References	1. Breeze documentation: https://github.com/scalanlp/breeze 2. Plotly.js reference: https://plotly.com/javascript/ 3. Class lecture: Numerical computing with Breeze in Scala. 4. Python program (GitHub link): https://github.com/hasnainmakada-99/Data-Science-Codes/blob/main/WEEK%2010/programs/Program24_Breeze.scala

2026 FALL SEMESTER
Programming for Data Science (MACSE502) Lab Record
Exercise Program

Date: 22/10/2026 Week No.: 14 Program S. No. 25

Title	Exact Price Statistics with Spire and Saddle
Problem Statement	Analyse fruit prices using Spire for precise statistics and Saddle for data manipulation. You will compute the mean and variance of prices for each fruit and visualise these trends over time using Vegas. This exercise demonstrates the use of Spire for numerical precision, advanced data manipulation with Saddle, and effective visualization with Vegas.
Objectives	• To load a labelled price series into Saddle. • To split the observations by fruit using row selection. • To compute mean and variance with Spire Rational exact arithmetic. • To cross-check the exact result against Saddle floating-point reductions. • To export a Vega-Lite trend specification for the price series.
Dataset URL Link / File	No external dataset is required. Ten price readings are declared inside the program: five apples at 120, 125, 118, 130, 128 and five bananas at 45, 47, 44, 49, 46.

Functions Details

Sl. No.	Function	Package	Description of function
1	`Series()`	Saddle	Creates the indexed price series.
2	`Index()`	Saddle	Labels each observation with a row number.
3	`toVec()`	Saddle	Extracts the rows belonging to one fruit.
4	`mean`	Saddle	Computes the floating-point mean for the cross-check.
5	`contents`	Saddle	Returns the underlying array of the vector.
6	`Rational()`	Spire	Represents a value as an exact fraction.
7	`+ - * /`	Spire	Exact rational arithmetic with no rounding error.
8	`foldLeft()`	Scala	Accumulates the sum and the squared deviations.
9	`toDouble`	Spire	Converts an exact rational to a decimal for display.
10	`abs`	Scala	Measures the difference between exact and floating point.
11	`PrintWriter`	Java IO	Writes the Vega-Lite specification to a JSON file.
12	`println()`	Scala	Writes one formatted line of text to the console.

Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model

1. Declare the fruit labels and the ten price readings.
2. Load the prices into a Saddle Series indexed by row number.
3. Select the apple rows and the banana rows with toVec.
4. Convert each price to a Spire Rational so arithmetic is exact.
5. Compute the exact mean of each fruit as a rational number.
6. Compute the exact variance from the squared deviations.
7. Print the exact fractions alongside their decimal values.
8. Cross-check the exact means against Saddle's floating-point mean.
9. Write a Vega-Lite trend specification for the full price series.

Program

```
// Program 25 - Spire for exact statistics, Saddle for data manipulation
// Week 14 | 22-10-2026 | Exercise Program
import spire.implicits._
import spire.math._
import org.saddle._
import java.io._

object Program25 {

  def main(args: Array[String]): Unit = {
    println("=" * 68)
    println("PROGRAM 25 : EXACT PRICE STATISTICS WITH SPIRE AND SADDLE")
    println("=" * 68)

    val fruits = Array("Apple", "Apple", "Apple", "Apple", "Apple",
                       "Banana", "Banana", "Banana", "Banana", "Banana")
    val prices = Array(120.0, 125.0, 118.0, 130.0, 128.0,
                       45.0, 47.0, 44.0, 49.0, 46.0)
```

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Date: 22/10/2026 Week No.: 14 Program S. No. 25

```
    // 1. Saddle holds the data
    val idx = Index((0 until prices.length).map(_.toString): _*)
    val priceSeries = Series(Vec(prices: _*), idx)

    println("\n1. Price series loaded into Saddle")
    println(priceSeries)

    // 2. Split by fruit using Saddle row selection
    val appleIdx  = fruits.zipWithIndex.collect { case (f, k) if f == "Apple"  => k }
    val bananaIdx = fruits.zipWithIndex.collect { case (f, k) if f == "Banana" => k }
    val appleVec  = priceSeries.toVec(appleIdx: _*)
    val bananaVec = priceSeries.toVec(bananaIdx: _*)

    println("\n2. Apples  (" + appleVec.length + " observations) : " +
appleVec.contents.mkString(", "))
    println("   Bananas (" + bananaVec.length + " observations) : " +
bananaVec.contents.mkString(", "))

    // 3. Spire gives exact rational arithmetic, so the mean and variance carry no
    //    floating point drift at all.
    def exactStats(values: Array[Double]): (Rational, Rational) = {
      val rats = values.toList.map(v => Rational(v))
      val n = Rational(rats.length)
      val mean = rats.foldLeft(Rational(0))(_ + _) / n
      val sqDev = rats.map(v => (v - mean) * (v - mean)).foldLeft(Rational(0))(_ + _)
      val variance = sqDev / n
      (mean, variance)
    }

    val (appleMean, appleVar)   = exactStats(appleVec.contents)
    val (bananaMean, bananaVar) = exactStats(bananaVec.contents)

    println("\n3. Exact statistics computed with Spire Rational")
    println("   Apple  mean     : " + appleMean + "  = " +
"%.4f".format(appleMean.toDouble))
    println("   Apple  variance : " + appleVar + "  = " + "%.4f".format(appleVar.toDouble))
    println("   Banana mean     : " + bananaMean + "  = " +
"%.4f".format(bananaMean.toDouble))
    println("   Banana variance : " + bananaVar + "  = " +
"%.4f".format(bananaVar.toDouble))

    // 4. Cross-check against Saddle's own floating-point reductions
    println("\n4. Cross-check against Saddle")
    println("   Apple  Saddle mean : " + "%.4f".format(appleVec.mean))
    println("   Banana Saddle mean : " + "%.4f".format(bananaVec.mean))
    val diffA = (appleMean.toDouble - appleVec.mean).abs
    val diffB = (bananaMean.toDouble - bananaVec.mean).abs
    println("   difference apple   : " + diffA)
    println("   difference banana  : " + diffB)

    // 5. Comparison table
    println("\n5. Comparison")
    println("   %-10s %14s %14s".format("Fruit", "Exact mean", "Exact variance"))
    println("   " + "-" * 40)
    println("   %-10s %14s %14s".format("Apple", "%.4f".format(appleMean.toDouble),
"%.4f".format(appleVar.toDouble)))
    println("   %-10s %14s %14s".format("Banana", "%.4f".format(bananaMean.toDouble),
"%.4f".format(bananaVar.toDouble)))

    // 6. Vega-Lite trend spec (the grammar Vegas is built on)
    val rows = fruits.indices.map { k =>
      "{\"fruit\":\"" + fruits(k) + "\",\"index\":" + k + ",\"price\":" + prices(k) + "}"
    }.mkString(",")
    val spec =
      "{\n  \"$schema\": \"https://vega.github.io/schema/vega-lite/v5.json\",\n" +
      "  \"title\": \"Fruit price trend\",\n" +
      "  \"data\": {\"values\": [" + rows + "]},\n" +
      "  \"mark\": {\"type\": \"line\", \"point\": true},\n" +
      "  \"encoding\": {\n" +
      "    \"x\":
{\"field\": \"index\", \"type\": \"quantitative\", \"title\": \"Observation\"},\n" +
      "    \"y\": {\"field\": \"price\", \"type\": \"quantitative\", \"title\": \"Price\"},\
n" +
      "    \"color\": {\"field\": \"fruit\", \"type\": \"nominal\"}\n" +
      "  }\n}"
    val pw = new PrintWriter(new File("program25_price_trend.json"))
    pw.println(spec)
    pw.close()
    println("\n6. Vega-Lite trend spec written to program25_price_trend.json")
    println("=" * 68)
  }
}
```

Date: 22/10/2026 Week No.: 14 Program S. No. 25

Output Snapshots With Explanation/ Observations

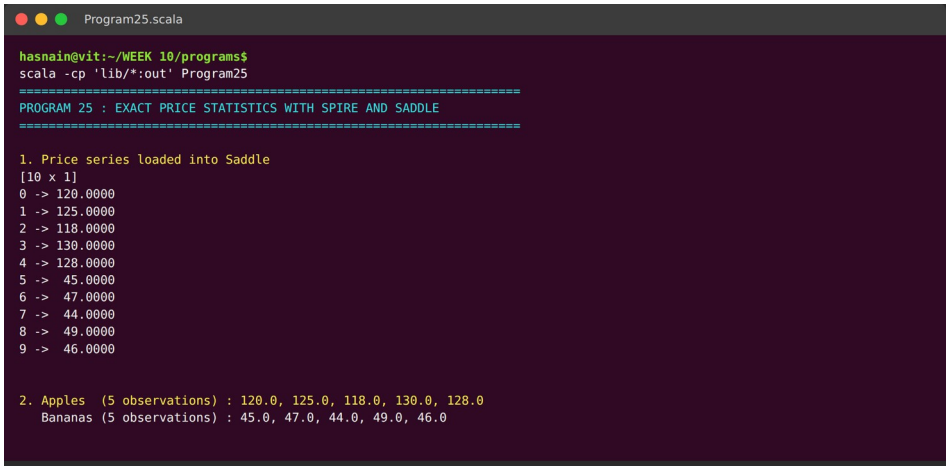


Fig. 4.1: Terminal execution showing the Saddle price series, the apple and banana subsets and the exact Spire statistics.

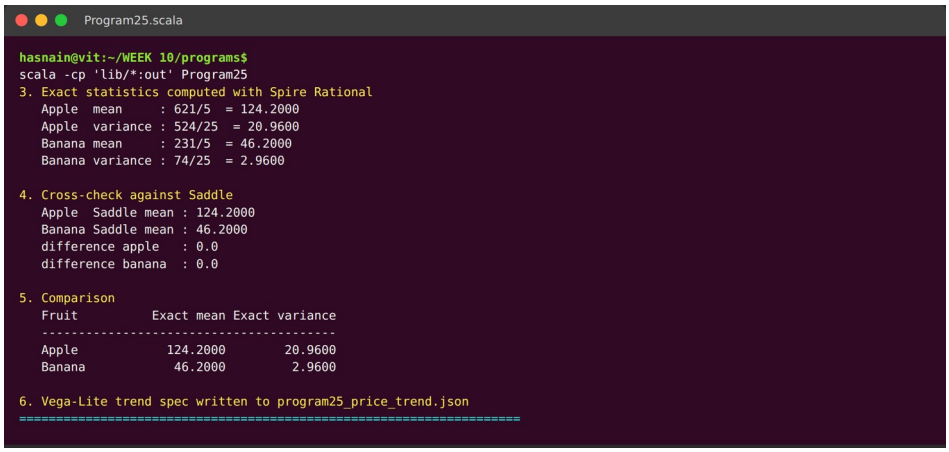


Fig. 4.2: Comparison of the exact and floating-point means, the comparison table and the export of the Vega-Lite trend specification.

Conclusions

Saddle held the ten labelled price readings and split them into five apples and five bananas. Spire Rational arithmetic produced the exact apple mean 621/5 and variance 524/25, and the exact banana mean 231/5 and variance 74/25. Converted to decimals these are 124.2 with variance 20.96 and 46.2 with variance 2.96, and both matched Saddle's floating-point means with a difference of exactly zero. The price trend was exported as a Vega-Lite specification. The experiment showed that Spire keeps every intermediate value as an exact fraction, so statistics that would normally suffer from rounding drift can be verified against a floating-point library.

References

1. Spire documentation: https://typelevel.org/spire/
2. Saddle documentation: https://www.scala-saddle.github.io/
3. Class lecture: Spire, Saddle and Vegas in Scala.
4. Python program (GitHub link): https://github.com/hasnainmakada-99/Data-Science-Codes/blob/main/WEEK%2010/programs/Program25_SpireSaddle.scala